Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD analytical/materials chemist with 6+ years in multi-technique characterization, method-development documentation, stability/degradation analysis, and DOE/QbD-style process readouts. Brings separations-science exposure (GPC/SEC, LC-MS-adjacent workflows) and technical-report depth to analytical method development while bridging honestly into biologics-specific platforms.

CORE COMPETENCIES

Analytical Method Development | Separations Science Bridge | CQA / QbD / DOE | Stability & Release Testing | Method Transfer Readiness | Technical Reports | Regulated Documentation Mindset | Cross-Functional CMC

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided analytical readout design for bio-based polymer process development, defining process windows, reproducibility criteria, and SOP-ready specifications.
- Connect FT-IR, thermal-analysis, spectroscopy, and process data to stability interpretation, troubleshooting, and technical recommendations.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Characterized 15+ functional material variants using UV-Vis-NIR, FT-IR, cyclic voltammetry, spectroelectrochemistry, NMR, GPC/SEC, TGA, DSC, and MS-informed analysis.
- Mapped degradation and stability failure modes through hypothesis-driven experiments, orthogonal characterization, structured datasets, and publication-quality technical documentation.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Developed molecular thin-film coating processes and linked CVD/surface-preparation variables with morphology, interface quality, and material performance.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Optimized CVD polymer thin-film deposition with DOE strategies and in-situ QCM-D monitoring to quantify growth kinetics and improve reproducibility.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30% and strengthened data integrity.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Analytical Method Development Bridge
Merck AR&D Match

Built analytical readouts and reproducibility criteria that connect material attributes to specifications, stability interpretation, and technical reports.

GPC/SEC, LC-MS exposure, FT-IR, UV-Vis-NIR, DOE/QbD

Stability and Degradation Characterization
Chemistry of Materials 2025

Mapped material stability and degradation mechanisms using orthogonal spectroscopy, electrochemistry, and thermal analysis.

FT-IR, UV-Vis-NIR, CV, DSC/TGA, NMR, GPC, MS

Process-Property Method Development
Current

Defined process windows and analytical readouts that support repeatable material quality and technical decision-making.

DOE, SPC, SOPs, technical reports

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Analytical Characterization: GPC/SEC, LC-MS and GC-MS exposure, UV-Vis-NIR, FT-IR, NMR, QCM-D, TGA, DSC, cyclic voltammetry, spectroelectrochemistry; UPLC/HPLC/CE biologics methods to be bridged

Method Development & Quality Mindset: DOE, QbD/SPC, method comparison, reproducibility studies, SOPs, technical reports, transfer-readiness documentation, data integrity

Stability & Materials Science: Material degradation, stability testing, process-property relationships, polymers, thin films, organic/materials chemistry

Communication: Publications, conference presentations, cross-functional technical summaries, lab notebooks, mentoring and training

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)